British Math Olympiad 2023 Round 1 — P2/6

Jonathan Kasongo

August 23, 2025

British Math Olympiad 2023 Round 1 — P2/6

The sequence of integers $a_0, a_1, \ldots$ has the property that for each $i \geq 2$, a_i is either $2a_{i-1} - a_{i-2}$ or $2a_{i-2} - a_{i-1}$.

Given that a_{2023} and a_{2024} are consecutive integers, prove that a_0 and a_1 are consecutive integers.

(Note that 6 and 7 are consecutive integers, as are 7 and 6)

Solution

Let n be a positive integer. Suppose that a_n and a_{n+1} are consecutive integers. That means that $|a_n - a_{n+1}| = 1$. But now depending on the defintion of a_{n+1} we have either $|a_n - (2a_n - a_{n-1})| = |a_{n-1} - a_n| = 1$ or $|a_n - (2a_{n-1} - a_n)| = 2|a_n - a_{n-1}| = 1$. Notice that the second case is impossible because it implies that there are two integers that differ by $\frac{1}{2}$. Thus we know that $|a_{n-1} - a_n| = 1$, that is, the two integers before a_n and a_{n+1} are also consecutive.

Now since we are given that a_{2023} and a_{2024} are consecutive integers, this implies that a_{2023} and a_{2022} are consecutive, and after repeating this arguement 2023 times we deduce that a_0 and a_1 are consecutive. $\blacksquare$